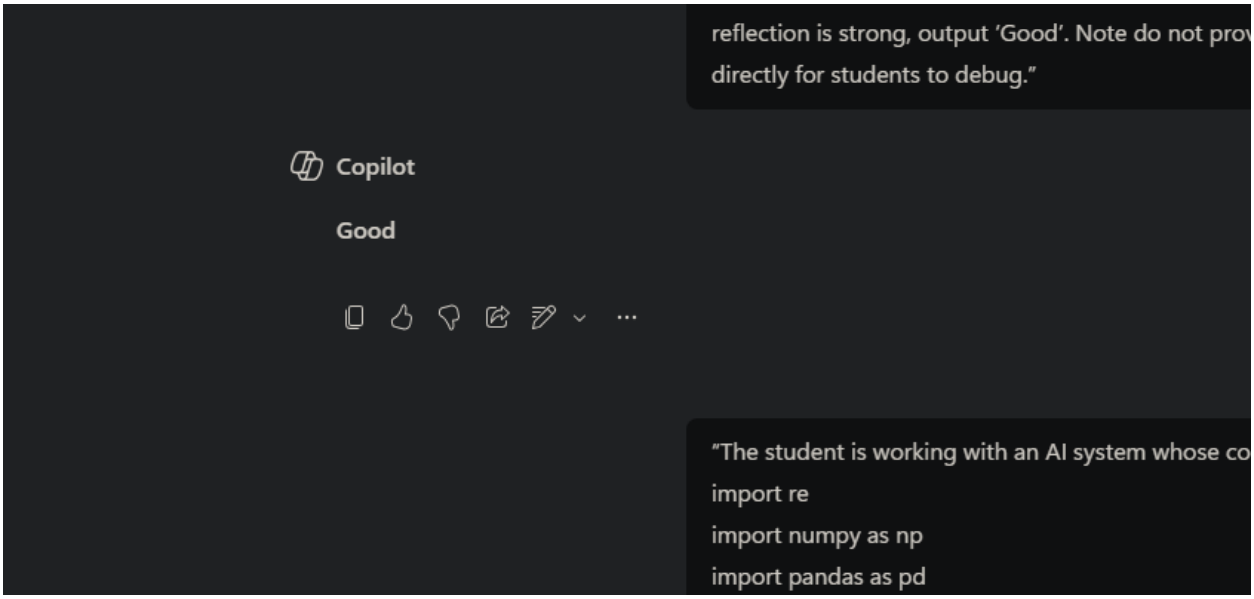
If I include my input this document will be 10,000 words long so I will only include copilot replies

1.



Error message:

:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-18 21:15:18.344666: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-18 21:15:19.534914: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Traceback (most recent call last): File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 249, in <module> main() File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 164, in main raise FileNotFoundError(FileNotFoundError: Can't find 'whateveer.xlsx'. Put the Excel file next to this script.

2.

Copilot

Bad

The reflection does **not** clearly identify the core issue, and the reasoning is vague and partially incorrect. It focuses on "distorted results" in a general way but doesn't precisely connect the behavior to how the program determines which labels are actually used during training.

Socratic guidance (to encourage deeper thinking)

Consider reflecting on questions like these:

- When you changed `LABEL_COL`, what parts of the code rely on that variable **by name** rather than by behavior?
- How does the program decide whether to use human-provided labels versus automatically generated ones?
- What evidence in the console output tells you **which labels were actually used** during training?
- The model trains and runs without errors—what does that suggest about whether the program *noticed* your change?
- How might silently falling back to a default path affect evaluation metrics such as precision, recall, and the confusion matrix?

Try tracing the data flow step by step—from loading the CSV, to selecting labels, to producing predictions—and ask yourself where your change *should* have had an effect but didn't.

I changed the label line, so that manual labels are not properly recognized. The code still ran thanks to the fallback mechanism in the line df["suggested_label"] = df[TEXT_COL].apply(heuristic_label). This means that even if the manual labels are missing or incorrect. The system will automatically create labels base on context.Which in the end cause the code to still ran, but the training behavior may differ and instead of using expected labels, it will use generated labels.is this good

Copilot

Good

Yes—this revised reflection correctly identifies the **core issue** and shows **deeper reasoning**:

- You recognized that changing `LABEL_COL` prevented the program from detecting manually provided labels.
- You correctly identified the **fallback mechanism** (`suggested_label`) as the reason the code continued to run without crashing.
- You explained the **consequence**: the model trained on automatically generated (heuristic) labels instead of the intended manual ones, which can significantly change training behavior and evaluation results.
- You distinguished between *program correctness* (it runs) and *semantic correctness* (it trains on the wrong labels), which is a key debugging insight.

This demonstrates an understanding not just of *what happened*, but *why* it happened and *why it matters*.

Error message:

2026-04-18 21:34:25.519672: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-18 21:34:26.743446: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-18 21:34:27.391009: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.4600 - loss: 1.0355 - val_accuracy: 0.5789 - val_loss: 0.9393 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.4933 - loss: 0.9508 - val_accuracy: 0.5789 - val_loss: 0.9551 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5333 - loss: 0.9065 - val_accuracy: 0.5789 - val_loss: 0.9619 Epoch 4/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.4400 - loss: 0.9647 - val_accuracy: 0.5789 - val_loss: 0.9509 Test Accuracy: 0.5106 Test Loss: 0.9563 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png.

3.

Copilot

Good

The reflection correctly identifies the **core issue** and demonstrates sound reasoning:

- It recognizes that the code still runs but learning degrades.
- It correctly explains the effect of an extremely high dropout rate on training (most information being discarded).
- It connects the hyperparameter change to poor model learning and low accuracy.
- It interprets what `0.9` means in the context of dropout and why that harms training.

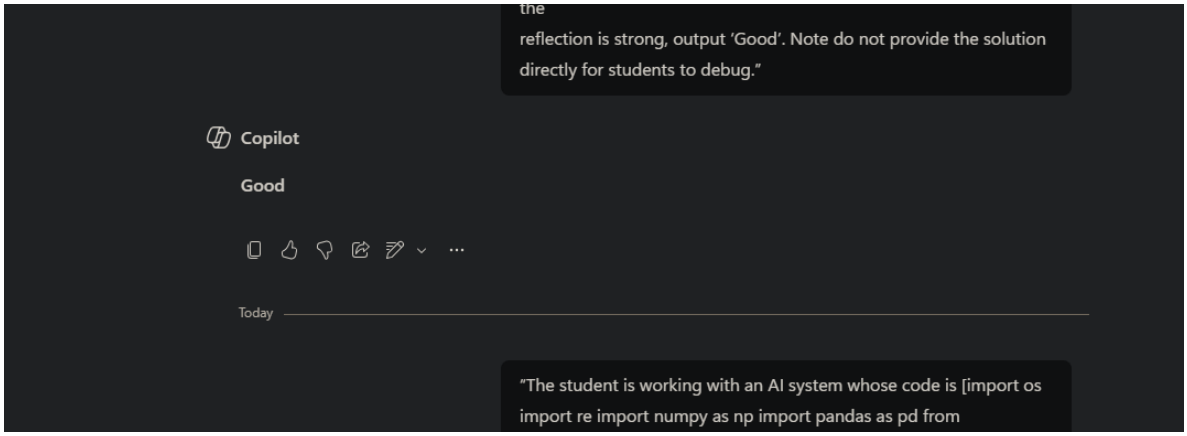
Overall, the student understands both *what* happened and *why*, rather than just describing surface-level symptoms.

Error message:

PS C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-18 23:33:28.390709: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-18 23:33:31.765039: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-18 23:33:32.919040: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.3867 - loss: 1.0735 - val_accuracy: 0.5789 - val_loss: 1.0085 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5000 - loss: 1.0156 - val_accuracy: 0.5789 - val_loss: 0.9681 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4667 - loss: 1.0146 - val_accuracy: 0.5789 - val_loss: 0.9592 Epoch 4/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4200 - loss: 1.0199 - val_accuracy: 0.5789 - val_loss: 0.9700 Test Accuracy: 0.5106 Test Loss: 1.0133 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-

defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png

4.



Error: message:

C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-18 23:33:28.390709: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-18 23:33:31.765039: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-18 23:33:32.919040: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow

binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.3867 - loss: 1.0735 - val_accuracy: 0.5789 - val_loss: 1.0085 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5000 - loss: 1.0156 - val_accuracy: 0.5789 - val_loss: 0.9681 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4667 - loss: 1.0146 - val_accuracy: 0.5789 - val_loss: 0.9592 Epoch 4/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4200 - loss: 1.0199 - val_accuracy: 0.5789 - val_loss: 0.9700 Test Accuracy: 0.5106 Test Loss: 1.0133 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png PS C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-18 23:47:39.952448: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-18 23:47:41.119409: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-18 23:47:41.767477: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.5333 - loss: 1.0232 - val_accuracy: 0.4474 - val_loss: 0.9497 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5333 - loss: 0.9476 - val_accuracy: 0.4474 - val_loss: 0.9204 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5333 - loss: 0.9619 - val_accuracy: 0.4474 - val_loss: 0.9101 Epoch 4/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5333 - loss: 0.9477 - val_accuracy: 0.4474 - val_loss: 0.9080 Test Accuracy: 0.5532 Test Loss: 0.9319 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.

_warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
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5.

Copilot

Good

The reflection correctly identifies the core issue and shows sound reasoning:

- It recognizes that changing the label from 1 to 8 caused a **hard crash during training**, not just degraded performance.
- It correctly connects the error to how `sparse_categorical_crossentropy` works—expecting class labels to be **integer indices within a fixed range**.
- It interprets the error message ("outside the valid range of [0, 3)") and links it to the model's number of output classes.
- It focuses on the **semantic mismatch between labels and model configuration**, rather than surface-level symptoms.

Overall, the student demonstrates an understanding of both *why* the error occurred and *how* the loss function enforces constraints on labels.

Error message:

[PS C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 00:04:54.566122: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 00:04:55.728134: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {8: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-19 00:04:56.362598: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following

instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 2026-04-19 00:04:56.902400: W tensorflow/core/framework/op_kernel.cc:1855] OP_REQUIRES failed at sparse_xent_op.cc:103 : INVALID_ARGUMENT: Received a label value of 8 which is outside the valid range of [0, 3). Label values: 8 8 8 0 8 8 0 0 8 8 0 0 8 8 8 0 2026-04-19 00:04:56.902593: I tensorflow/core/framework/local_rendezvous.cc:407] Local rendezvous is aborting with status: INVALID_ARGUMENT: Received a label value of 8 which is outside the valid range of [0, 3). Label values: 8 8 8 0 8 8 0 0 8 8 0 0 8 8 8 0 [[{{function_node __inference_one_step_on_data_1475}}{{node compile_loss/sparse_categorical_crossentropy/SparseSoftmaxCrossEntropyWithLogits/SparseSoftmaxCrossEntropyWithLogits}}]] Traceback (most recent call last): File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 249, in <module> main() File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 202, in main history = model.fit(File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\utils\traceback_utils.py", line 122, in error_handler raise e.with_traceback(filtered_tb) from None File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\tensorflow\python\eager\execute.py", line 53, in quick_execute tensors = pywrap_tfe.TFE_Py_Execute(ctx._handle, device_name, op_name, tensorflow.python.framework.errors_impl.InvalidArgumentError: Graph execution error: Detected at node compile_loss/sparse_categorical_crossentropy/SparseSoftmaxCrossEntropyWithLogits/SparseSoftmaxCrossEntropyWithLogits defined at (most recent call last): File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 249, in <module> File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 202, in main File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\utils\traceback_utils.py", line 117, in error_handler File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 399, in fit File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 241, in function File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 154, in multi_step_on_iterator File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 125, in wrapper File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 134, in one_step_on_data File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\trainer.py", line 62, in train_step File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\trainers\trainer.py", line 383, in _compute_loss File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\trainers\trainer.py", line 351, in compute_loss File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\trainers\compile_utils.py", line 699, in __call__ File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\trainers\compile_utils.py", line 724, in call File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\losses\loss.py", line 67, in __call__ File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\losses\losses.py", line 33, in call File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\losses\losses.py", line 2330, in sparse_categorical_crossentropy File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\ops\nn.py", line 2016, in sparse_categorical_crossentropy File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\nn.py", line 753, in sparse_categorical_crossentropy Received a label value of 8 which is outside the valid range of [0, 3). Label values: 8 8 8 0 8 8 0 0 8 8 0 0 8 8 8 0 [[{{node

compile_loss/sparse_categorical_crossentropy/SparseSoftmaxCrossEntropyWithLogits/SparseSoftmaxCrossEntropyWithLogits}}]] [Op:__inference_multi_step_on_iterator_1532]

6.

Copilot

Good

The reflection accurately identifies the core issue and demonstrates solid reasoning:

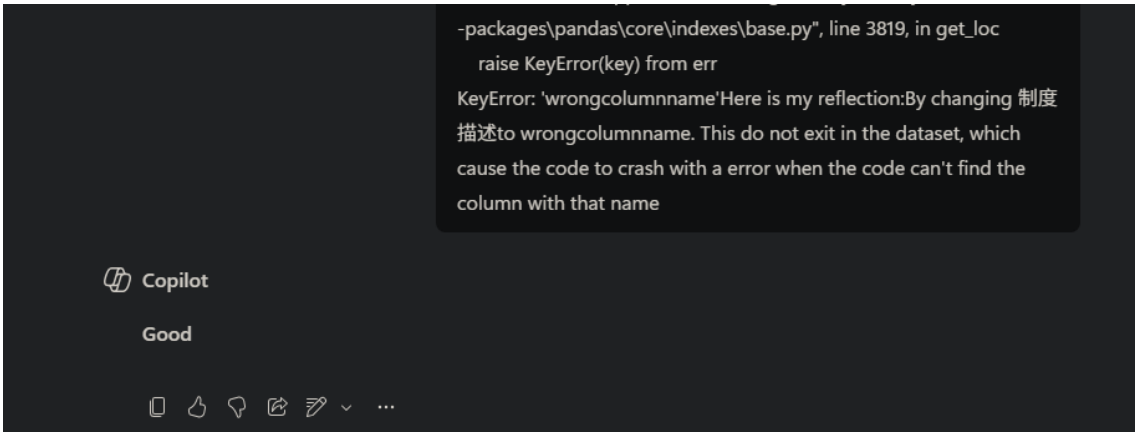
- It correctly recognizes that the model is configured for **multi-class prediction (3 classes)**.
- It correctly explains that switching to `binary_crossentropy` creates a **mismatch between the loss function's expectations and the model's output shape**.
- It links the error message (rank/shape mismatch) to the conceptual problem: binary loss functions expect binary-style targets, not multi-class outputs.
- The explanation focuses on **model–loss compatibility**, not just the surface-level crash.

Overall, the student shows a clear understanding of *why* the error occurred and how the change altered the assumptions of the training setup.

Error message:

PS C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 00:28:28.714934: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 00:28:29.934300: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-19 00:28:30.572465: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 Traceback (most recent call last): File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 249, in <module> main() File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 202, in main history = model.fit(File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\utils\traceback_utils.py", line 122, in error_handler raise e.with_traceback(filtered_tb) from None File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\backend\tensorflow\nn.py", line 776, in binary_crossentropy raise ValueError(ValueError: Arguments `target` and `output` must have the same rank (ndim). Received: target.shape=(None,), output.shape=(None,

7.



Error message:

PS C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 15:14:35.577033: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 15:14:38.623611: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Traceback (most recent call last): File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\indexes\base.py", line 3812, in get_loc return self._engine.get_loc(casted_key) File "pandas/_libs/index.pyx", line 167, in pandas._libs.index.IndexEngine.get_loc File "pandas/_libs/index.pyx", line 196, in pandas._libs.index.IndexEngine.get_loc File "pandas/_libs/hashtable_class_helper.pxi", line 7088, in pandas._libs.hashtable.PyObjectHashTable.get_item File "pandas/_libs/hashtable_class_helper.pxi", line 7096, in pandas._libs.hashtable.PyObjectHashTable.get_item KeyError: 'wrongcolumnname' The above exception was the direct cause of the following exception: Traceback (most recent call last): File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 249, in <module> main() File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 168, in main df = load_and_clean(EXCEL_PATH) File "C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project\train_996_text_classifier.py", line 49, in load_and_clean df = df[df[TEXT_COL].notna()] File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\frame.py", line 4113, in __getitem__ indexer = self.columns.get_loc(key) File "C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\indexes\base.py", line 3819, in get_loc raise KeyError(key) from err KeyError: 'wrongcolumnname'

8.

Copilot

Good

The reflection correctly identifies the core issue and shows appropriate reasoning:

- It recognizes that changing from a softmax layer with multiple outputs to a single sigmoid output fundamentally **changes the problem formulation**.
- It correctly notes that the model is no longer aligned with a **3-class classification task**, even though the code still runs.
- It distinguishes between *the program executing successfully* and *the model being conceptually correct*, which demonstrates deeper understanding.
- It explains the architectural mismatch (single output vs. multi-class labels) rather than focusing only on surface-level metrics.

Overall, the student shows awareness of how output layer design affects model semantics, not just runtime behavior.

Error message:

C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 00:40:24.908849: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 00:40:26.078728: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-19 00:40:26.720599: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.5200 - loss: 0.9935 - val_accuracy: 0.5789 - val_loss: 0.9487 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5200 - loss: 0.9233 - val_accuracy: 0.5789 - val_loss: 0.9609 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4867 - loss: 0.9449 - val_accuracy: 0.5789 - val_loss: 0.9655 Epoch 4/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5000 - loss: 0.9252 - val_accuracy: 0.5789 - val_loss: 0.9636 Test Accuracy: 0.5106 Test Loss: 0.9417 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-

packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png

9.

Copilot

Good

The reflection correctly identifies the core issue and demonstrates thoughtful reasoning:

- It recognizes that drastically reducing `VOCAB_SIZE` severely limits the text representation capacity of the model.
- It distinguishes between **successful execution** and **model effectiveness**, noting that the code can still run while learning quality degrades.
- It explains *why* learning is weakened (loss of expressive power in character embeddings), not just *that* results are poor.
- It avoids overclaiming about crashes or errors and instead focuses on representational limits and their impact.

Overall, this shows an understanding of how preprocessing and model capacity affect learning, rather than a superficial description of output metrics.

Error message:

C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 00:49:11.068831: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 00:49:12.206253: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank, training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-19 00:49:12.820109: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.4933 - loss: 0.9959 - val_accuracy: 0.5789 - val_loss: 0.9310 Epoch 2/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5067 - loss: 0.9346 - val_accuracy: 0.5789 - val_loss: 0.9454 Epoch 3/15 10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4667 - loss: 0.9463 - val_accuracy: 0.5789 - val_loss: 0.9545 Epoch 4/15 10/10 ━━━━

━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4667 - loss: 0.9381 - val_accuracy: 0.5789 - val_loss: 0.9384 Test Accuracy: 0.5106 Test Loss: 0.9443 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png

10

Copilot

Good

The reflection identifies the core issue and shows clear, non-superficial reasoning:

- It correctly explains why training still runs (batch size larger than dataset is allowed).
- It recognizes the **mechanical effect** (only one step per epoch).
- It explains the **learning consequence** (fewer parameter updates, weaker gradient signal).
- It distinguishes runtime success from **training quality and experimental validity**.

Overall, the student understands both the technical behavior and its implications for learning, rather than just reporting surface-level results.

Error message:

C:\Users\keke1\Documents\GitHub\AI Final\AI-100 final\996_project> python train_996_text_classifier.py 2026-04-19 00:57:27.233882: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. 2026-04-19 00:57:28.314496: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`. Loaded rows: 235 Columns: ['所在城市', '公司名字', '曝光/施行时间', '制度描述'] [Saved] to_label_996.csv Open it, fill the 'label' column with 0/1/2 (optional). If you leave 'label' blank,

training will use suggested_label. Num classes: 3 Class counts: {1: np.int64(123), 0: np.int64(88), 2: np.int64(24)} 2026-04-19 00:57:28.930705: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags. Epoch 1/15 1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 579ms/step - accuracy: 0.3800 - loss: 1.1013 - val_accuracy: 0.5789 - val_loss: 1.0421 Epoch 2/15 1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 54ms/step - accuracy: 0.4600 - loss: 1.0417 - val_accuracy: 0.5789 - val_loss: 1.0008 Epoch 3/15 1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 0.4800 - loss: 0.9928 - val_accuracy: 0.5789 - val_loss: 0.9715 Epoch 4/15 1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 0.5333 - loss: 0.9698 - val_accuracy: 0.5789 - val_loss: 0.9524 Test Accuracy: 0.5106 Test Loss: 1.0420 Classification report: C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) C:\Users\keke1\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior. _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0]) precision recall f1-score support 0 0.0000 0.0000 0.0000 18 1 0.5106 1.0000 0.6761 24 2 0.0000 0.0000 0.0000 5 accuracy 0.5106 47 macro avg 0.1702 0.3333 0.2254 47 weighted avg 0.2608 0.5106 0.3452 47 Confusion matrix: [[0 18 0] [0 24 0] [0 5 0]] WARNING:absl:The `save_format` argument is deprecated in Keras 3. We recommend removing this argument as it can be inferred from the file path. Received: save_format=h5 WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`. [Saved model] model_996_text_classifier.h5 [Saved] training_curve.png